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Have you ever thought about creating a song of your own? But you are still thinking about what makes a song popular on music streaming platforms? How can my song be in the Top 50 charts on Spotify? How can my song become a “hit song”? I am sure when you want to create a song of your own, these questions are probably in your head. Through this story, I will walk you through the factors that make songs on Spotify popular. For the visualization tool, I chose to use Power Bi.

In the first step, I will need to retrieve the data (you can do a Google search for Spotify data). Using Kaggle, I retrieved the dataset called “Top 50 Spotify Songs — 2019”, which can be found [here](#). After retrieving the dataset and analyzing it, there are a total of 50 songs which also include audio features such as energy, danceability, loudness, liveness, valence, acousticness, speechiness. Moreover, the dataset also includes track names, artist names, and popularity. And I can also access the genres of each song and find the top 5 genres for my dataset, which are: dance pop, pop, latin, Canadian hip hop, and edm.

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Song genres by popularity

Now let's have a look at each of the audio features:

Energy

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Energy by popularity

Let's first talk about what is energy?, energy is a measure from 0 to 100 and represents a perceptual measure of intensity and activity. Typically, energetic tracks feel fast, loud, and noisy. For example, death metal has high energy, while a Bach prelude scores low on the scale. Perceptual features contributing to this attribute include dynamic range, perceived loudness, timbre, onset rate, and general entropy. After understanding what is energy, look at the scatter plot above. It can be observed that songs with less than 40 and more than 80 energy are not very popular, but instead of creating a song that has energy ranging from 43 to 65, it can be popular.

Danceability

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Danceability by popularity

What is danceability? Danceability describes how suitable a track is for dancing based on a combination of musical elements including tempo, rhythm stability, beat strength, and overall regularity. A value of 0 is least danceable and 100 is most danceable. And in this line graph, it can be

seen that not all danceable songs are popular. Likewise, songs with a danceability of 64 to 84 are the most popular songs on Spotify.

Loudness

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The overall loudness of a track in decibels (dB). Loudness values are averaged across the entire track and are useful for comparing the relative loudness of tracks. Loudness is the quality of a sound that is the primary psychological correlate of physical strength (amplitude). Values typically range between -60 and 0 db. Then the question arrived, which loudness is best for the song? The graph clearly shows that different loudnesses can make a song popular, but other factors must be considered, as some songs with the same loudness as popular songs are not popular. In addition, loudness around -18 to -5 is very common in the Top 50 songs on Spotify.

Valence

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A measure from 0 to 100 describes the musical positiveness conveyed by a track. This scatter plot shows us that the valence doesn't have much effect on the popularity of the song. As you can see, almost all valences can become popular songs, even valence 0 can become popular.

Acousticness

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Does acousticness affect the popularity of the song? A confidence measure from 0 to 100 of whether the track is acoustic. 100 represents high confidence the track is acoustic. So, by looking at the graph, do you think having more acoustic or less acoustic is good? The answer is that having acousticness more than 60 is not a good idea, but between 1 and 59 is a good range of acousticness. But sometimes people also don't like the acousticness around 1 to 30 which means acousticness also depends on the **genre** of the song.

Speechnicess

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How does speechiness affect the popularity of the song? Speechiness detects the presence of spoken words in a track. Therefore, it is seen that the less spoken words there are, the more popular the song would be. Moreover, having a presence of spoken words makes your song less popular as people usually want to listen to the lyrics of the song, not the spoken words.

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In addition, I also found the average values for each audio feature: energy, danceability, loudness, valence, acousticness, speechness. These values

could be used to help you create a song that has a higher chance of being featured in the Top 50 Spotify songs in the future.

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Audio features average

Moreover, the artist of the song can also make the song itself popular. In the bar graph below, you can see that Ed Sheeran has 4 songs out of the Top 50 Spotify songs, and he has the highest popularity.

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Number of Song by Artists

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Artists' Popularity

Now you may be wondering which songs I am talking about? So let's look at the list of the songs.

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Top 50 Spotify songs

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Top 50 Spotify songs

To conclude, by using Power Bi to generate the visual, we can see that the successful songs or popular songs in the Top 50 are mostly from well-known artists like Ed Sheeran, Taylor Swift, Ariana Grande, The Chainsmokers, Marshmello, Lady Gaga, Khalid, DJ Snake, Ladt Gaga, and etc. Likewise, categories that the songs are usually in are dance pop, pop, latin, Canadian hip hop, and EDM. In addition, I also found average value of each audio

features that can help you to create a songs which have a chance of becoming Top 50 Spotify songs in the coming future. Dream it. Believe it. Build it.